Tutorial No : AMVS 3

Topic : DUCT SYSTEM & SIZING

1. State the purpose(s) of the following:

   (a) Wrapping a supply air duct with fiberglass and double sided aluminium foil. [3 points]
   (b) Internally insulating a main supply duct and return duct at the air handling unit. [3 points]

2. With the aid of a diagram, explain the terms “throw” and “drop” as applied to supply air outlets. [6 points]

3. List down five (5) design objectives and considerations. [5 points]

4. 4 Figure 4 shows the ductwork layout for a fan coil unit. Identify the index run. Size all sections of rectangular ductwork in the index run using the Equal Friction Method, Friction Loss Chart and Round to Rectangular Duct Conversion Table. Calculate the external fan static pressure required. [21 points]

   You shall use the following information:

   Duct design friction loss is 1.0 Pa/m.
   Loss coefficient at the centrifugal fan discharge is 0.7.
   Loss coefficients of all elbows and tees are shown in Figure 4.
   Pressure loss at each supply air diffuser and each return air grille is 20 Pa.
   Depths of ducts are to be maintained at a maximum of 300 mm.
   Discharge flow rate at each diffuser is 500 l/s.
   Return air flow rate at each return air grille is the same.
   Outdoor air flow rate is 20% of the total supply air flow rate.
   Safety factor is 20%.

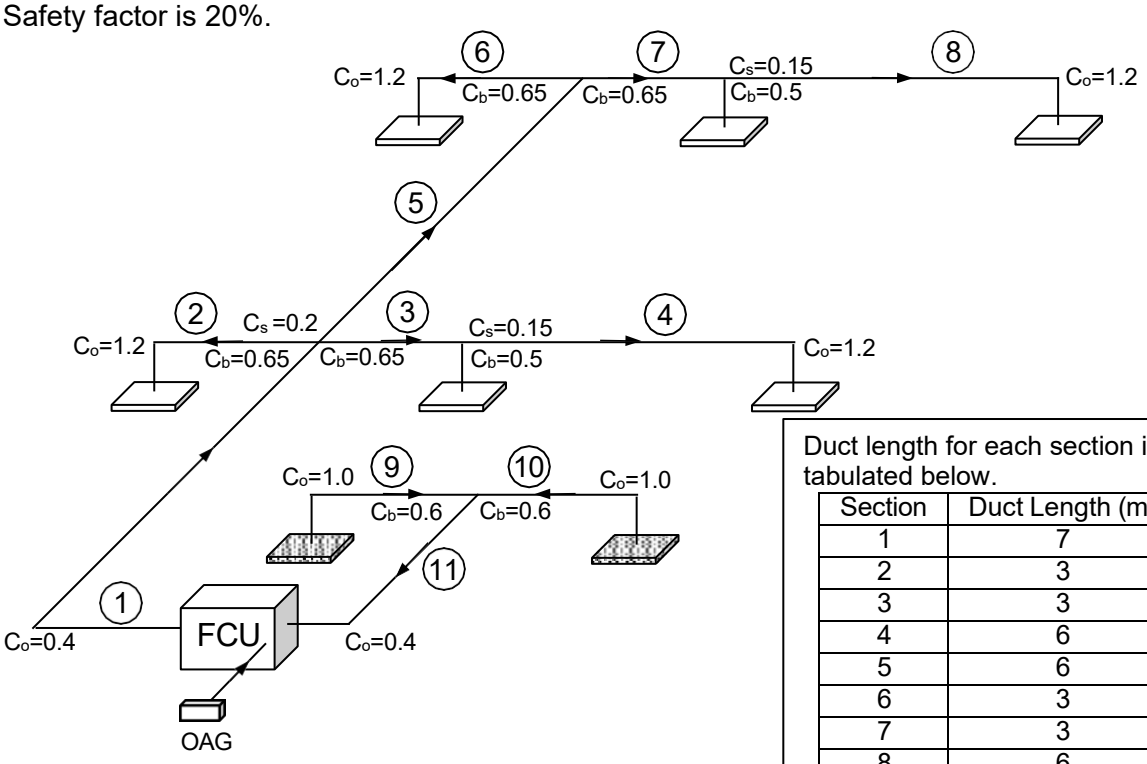


Figure 4

Duct length for each section is tabulated below.

Section	Duct Length (m)
1	7
2	3
3	3
4	6
5	6
6	3
7	3
8	6
9	3
10	4
11	3

5. Figure 5 shows the ductwork layout for an air handling unit serving an air-conditioned office. Identify the index run. Size all sections of rectangular ductwork in the index run using the Equal Friction Method, Friction Loss Chart and Round to Rectangular Duct Conversion Table. Calculate the external fan static pressure required. [24 points]

You may use the following assumptions:

- Main duct frictional loss is 0.8 Pa/m.
- Loss coefficient at centrifugal fan is 0.63
- Loss coefficients of all elbows and tees are indicated on Figure 5.
- Losses at supply air diffuser and return air grille are 20 Pa each.
- Depth of duct to be maintained at a maximum of 300 mm.
- Discharge flow rate at each diffuser is 500 l/s.
- Outdoor air flow rate is 15% of the total supply air flow rate.
- Safety factor is 20%.

5.

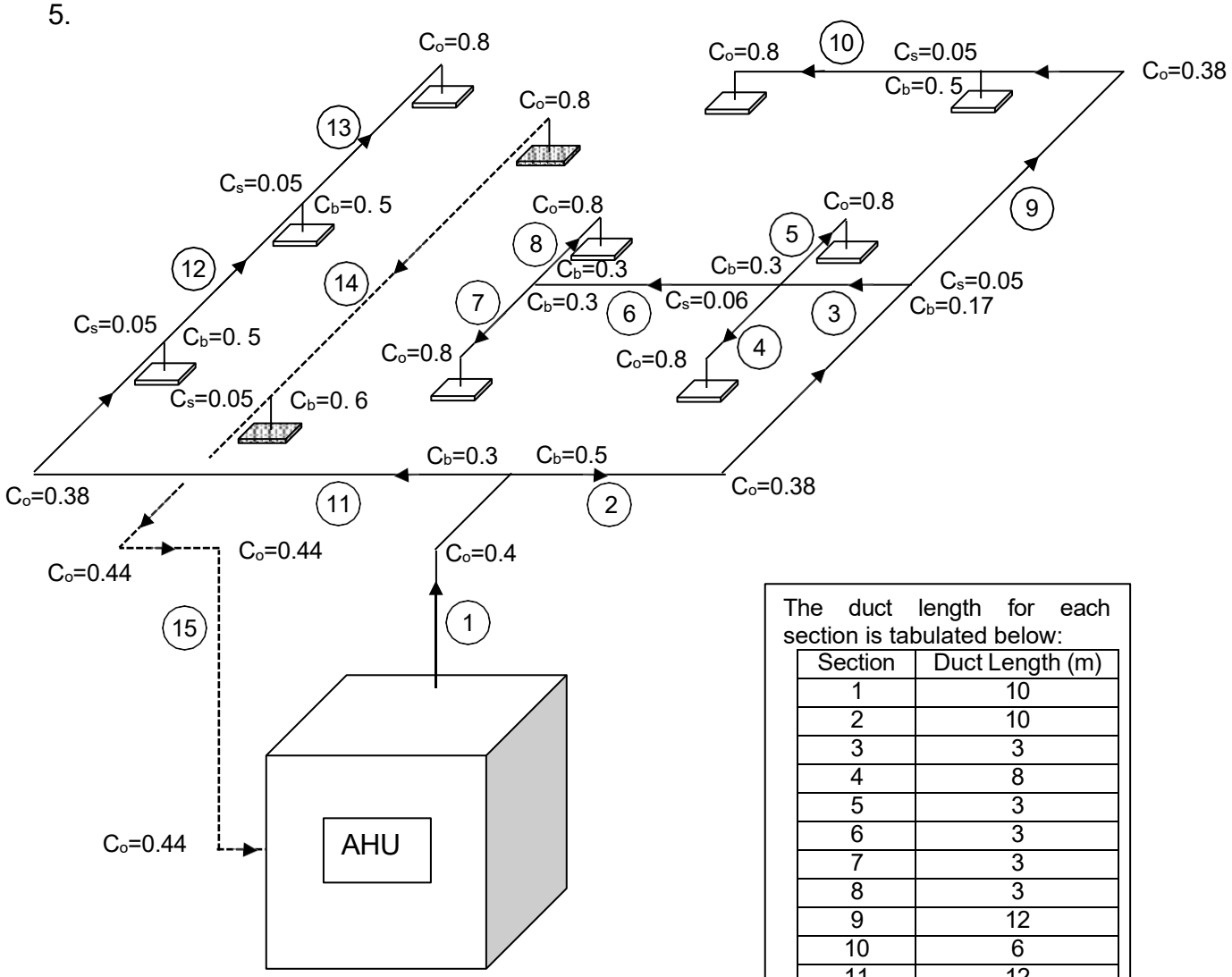


Figure 5

The duct length for each section is tabulated below:

Section	Duct Length (m)
1	10
2	10
3	3
4	8
5	3
6	3
7	3
8	3
9	12
10	6
11	12
12	6
13	6
14	10
15	13